radius. Once it reaches near the edge, it is taken out with the help of deflector plate and allowed to hit the target T.

Very important condition in cyclotron operation is the resonance condition. It happens when the frequency f at which the positive ion circulates in the magnetic field must be equal to the constant frequency of the electrical oscillator f_{osc}

From equation (3.59), we have

$$f_{OSC} = \frac{qB}{2\pi m}$$

The time period of oscillation is

$$T = \frac{2\pi m}{qB}$$

The kinetic energy of the charged particle is

$$KE = \frac{1}{2}mv^2 = \frac{q^2B^2r^2}{2m} \qquad (3.63)$$

Limitations of cyclotron

(a) the speed of the ion is limited
(b) electron cannot be accelerated
(c) uncharged particles cannot be accelerated

Note

Deutrons (bundles of one proton and one neutron) can be accelerated because it has same charge as that of proton. But neutron (electrically neutral particle) cannot be accelerated by the cyclotron. When a deutron is bombarded with a beryllium target, a beam of high energy neutrons are produced. These high-energy neutrons are sent into the patient's cancerous region to break the bonds in the DNA of the cancer cells (killing the cells). This is used in treatment of fast-neutron cancer therapy.

EXAMPLE 3.26

Suppose a cyclotron is operated to accelerate protons with a magnetic field of strength 1 T. Calculate the frequency in which the electric field between two Dees could be reversed.

Solution

Magnetic field B = 1 T
Mass of the proton, $m_p = 1.67\times10^{-27}\,kg$
Charge of the proton, $q = 1.60\times10^{-19}\,C$

$$f = \frac{qB}{2\pi m_p} = \frac{\left(1.60\times10^{-19}\right)(1)}{2(3.14)(1.67\times10^{-27})}$$

$$= 15.3\times10^{6}\,Hz = 15.3\,MHz$$

3.10.5 Force on a current carrying conductor placed in a magnetic field

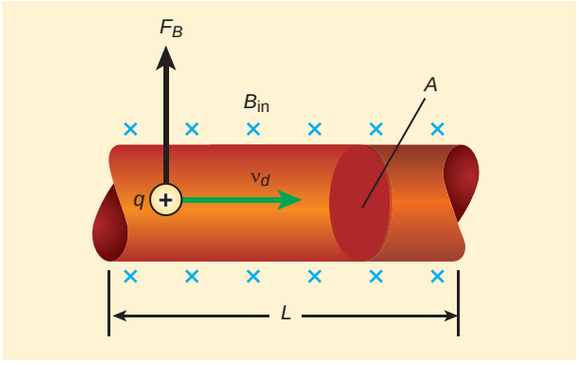


Figure 3.56 Current carrying conductor in a magnetic field

When a current carrying conductor is placed in a magnetic field, the force experienced by the wire is equal to the sum of Lorentz forces on the individual charge carriers in the wire. Consider a small segment of wire of length dl, with cross-sectional area A and current I as shown in Figure 3.56. The free electrons drift opposite to the direction of current. So the relation between current I and magnitude of drift velocity v_d (Refer Unit 2) is

$$I = neAvd \qquad (3.64)$$

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